

# NeuroX Adaptive

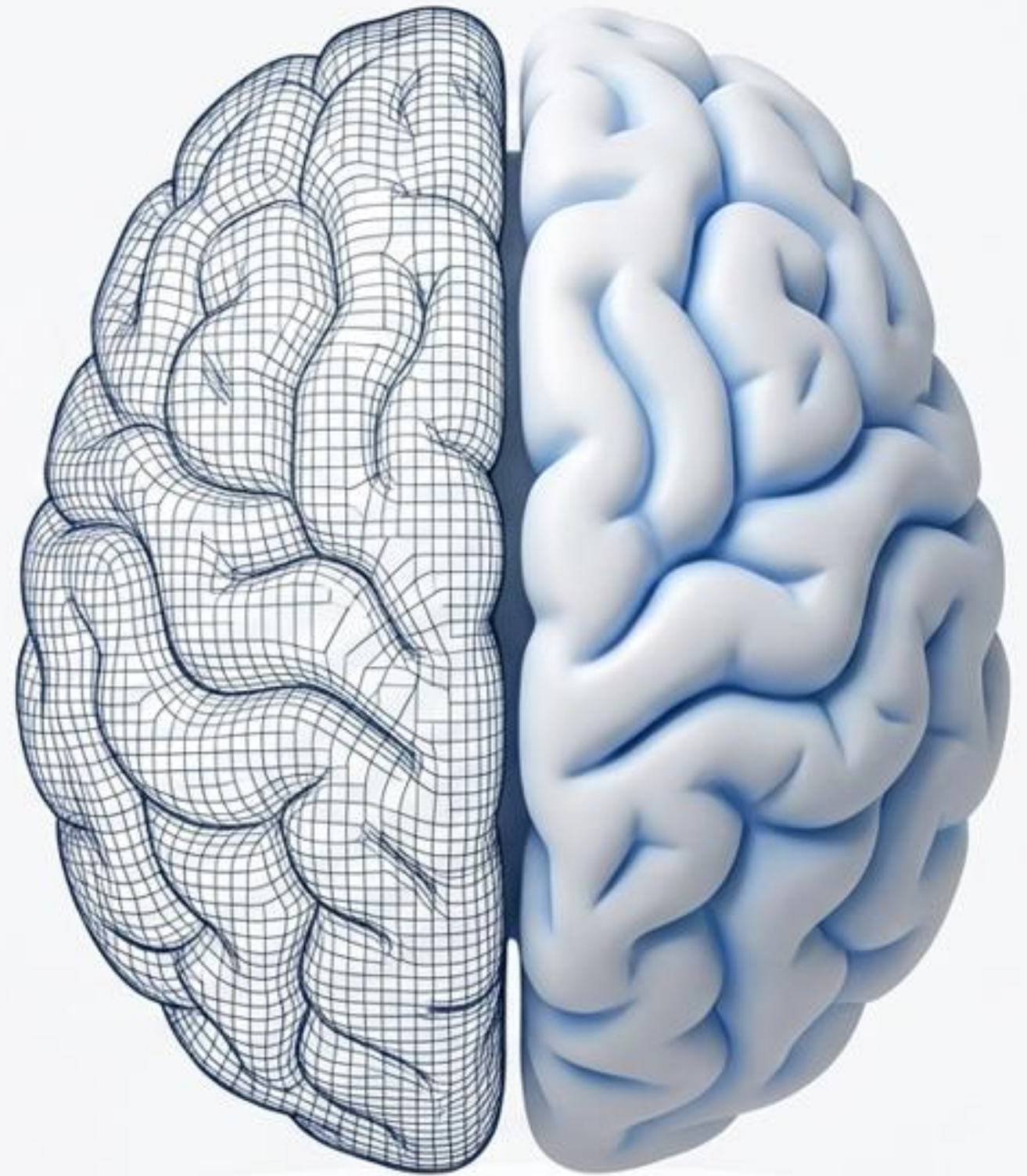
A Hybrid 3D CNN-Transformer  
Framework for Holistic  
Neuroimaging

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**Guide Name:**

1. Dr. Jayaram B

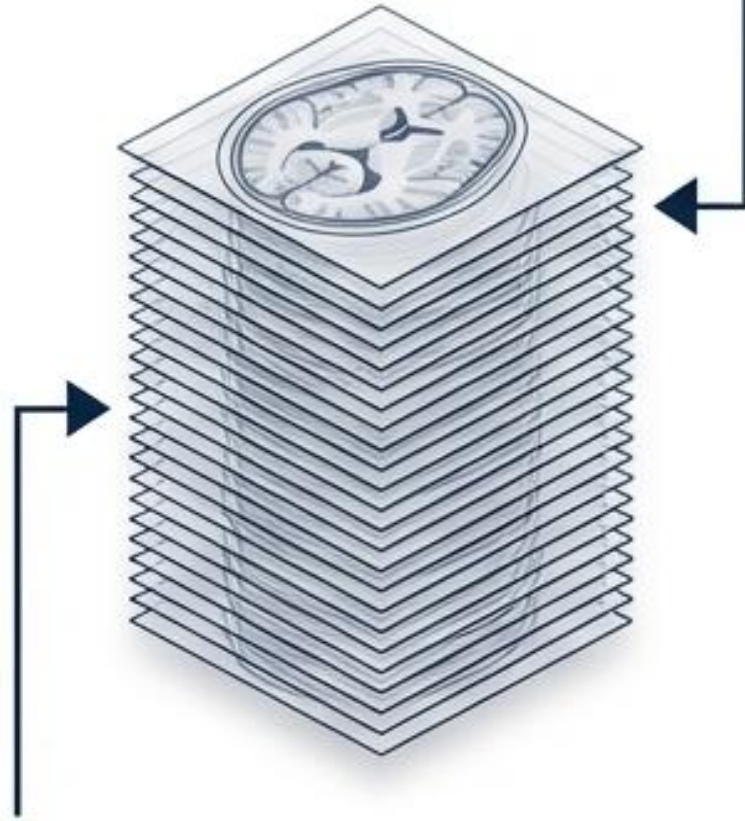


# Introduction

- **Current Diagnostic Bottleneck:** Modern medical imaging, such as MRI and CT, produces massive 3D volumetric datasets (NIfTI) that require exhaustive manual review by radiologists, leading to potential diagnostic fatigue and delays.
- **The Problem of "Siloed" AI:** Existing Computer-Aided Diagnosis (CAD) tools are typically task-specific, meaning a clinician must run separate models for tumor detection, stroke analysis, and neurodegeneration (Alzheimer's).
- **The Need for Volumetric Context:** Standard 2D slice-by-slice analysis often fails to capture the global structural symmetry and long-range anatomical relationships necessary for early-stage pathology detection.

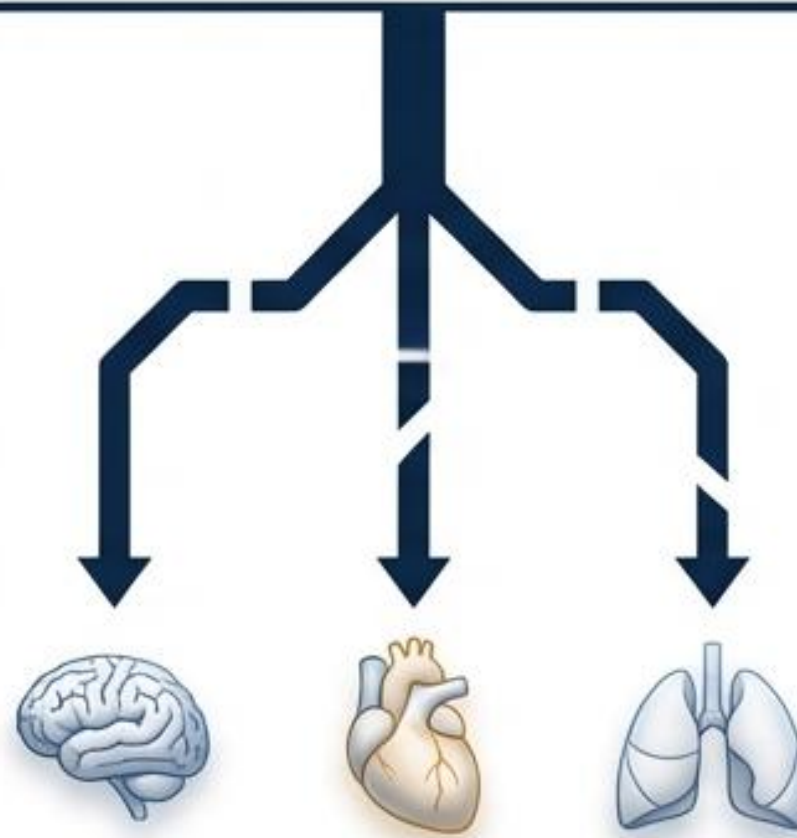


# The Clinical Paradigm & Problem Statement



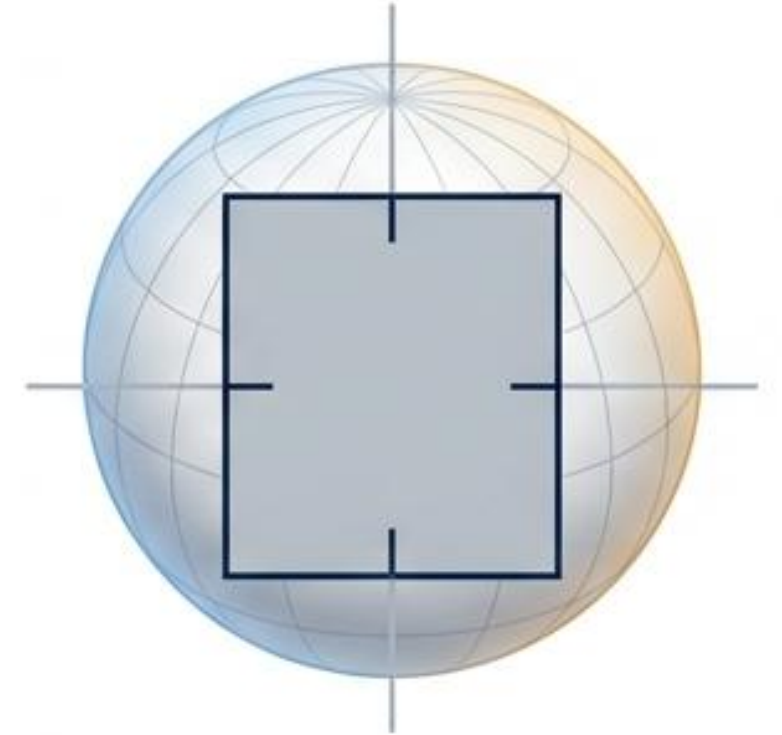
## The Overload

Modern scans produce **hundreds** of slices per patient, directly causing severe radiologist fatigue.



## The Fragmentation

Clinicians are forced into task-**specific** silos, running separate software for tumors, strokes, and Alzheimer's.



## The Blindspot

Single-slice 2D analysis fundamentally misses volumetric depth and systemic structural symmetry.

# Project Vision & Core Objectives

## NeuroX Adaptive Core System

```
graph TD; A[NeuroX Adaptive Core System] --> B[Unification]; A --> C[Accuracy]; A --> D[Explainability]; A --> E[Automation];
```

### Unification

1 Model, 1 NIfTI Volume. Simultaneous diagnosis of Brain Tumors, Stroke, and Alzheimer's.

### Accuracy

Targeting >85% baseline precision leveraging a unique Hybrid 3D CNN-Transformer architecture.

### Explainability

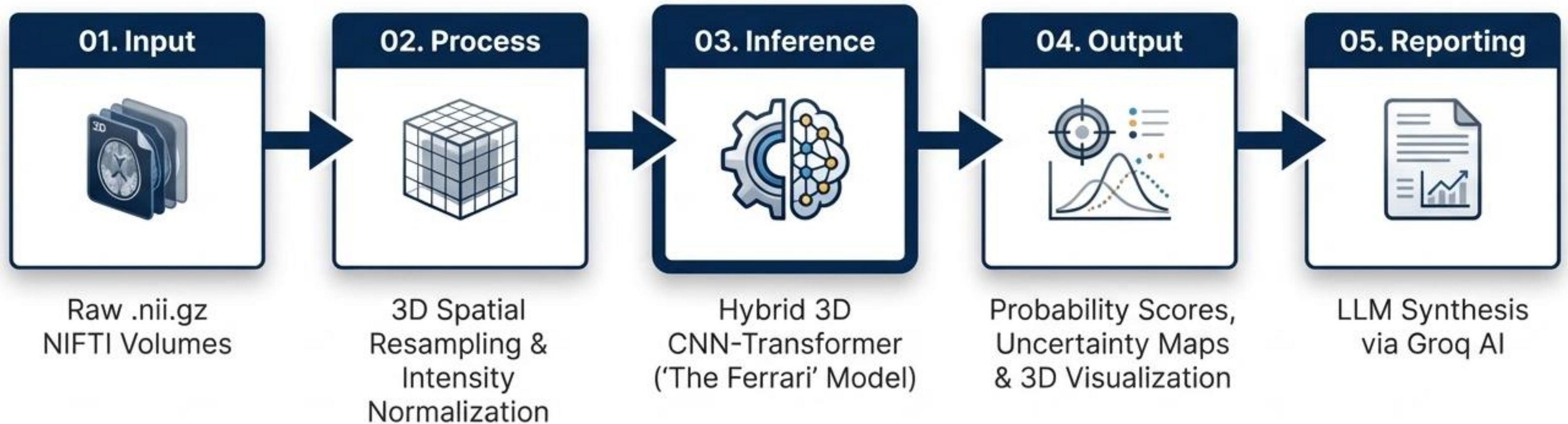
Bridging the physician trust gap with 3D Grad-CAM heatmaps for instant visual verification.

### Automation

Instant generation of structured radiology reports via Groq AI LLM synthesis.



# High-Level System Workflow

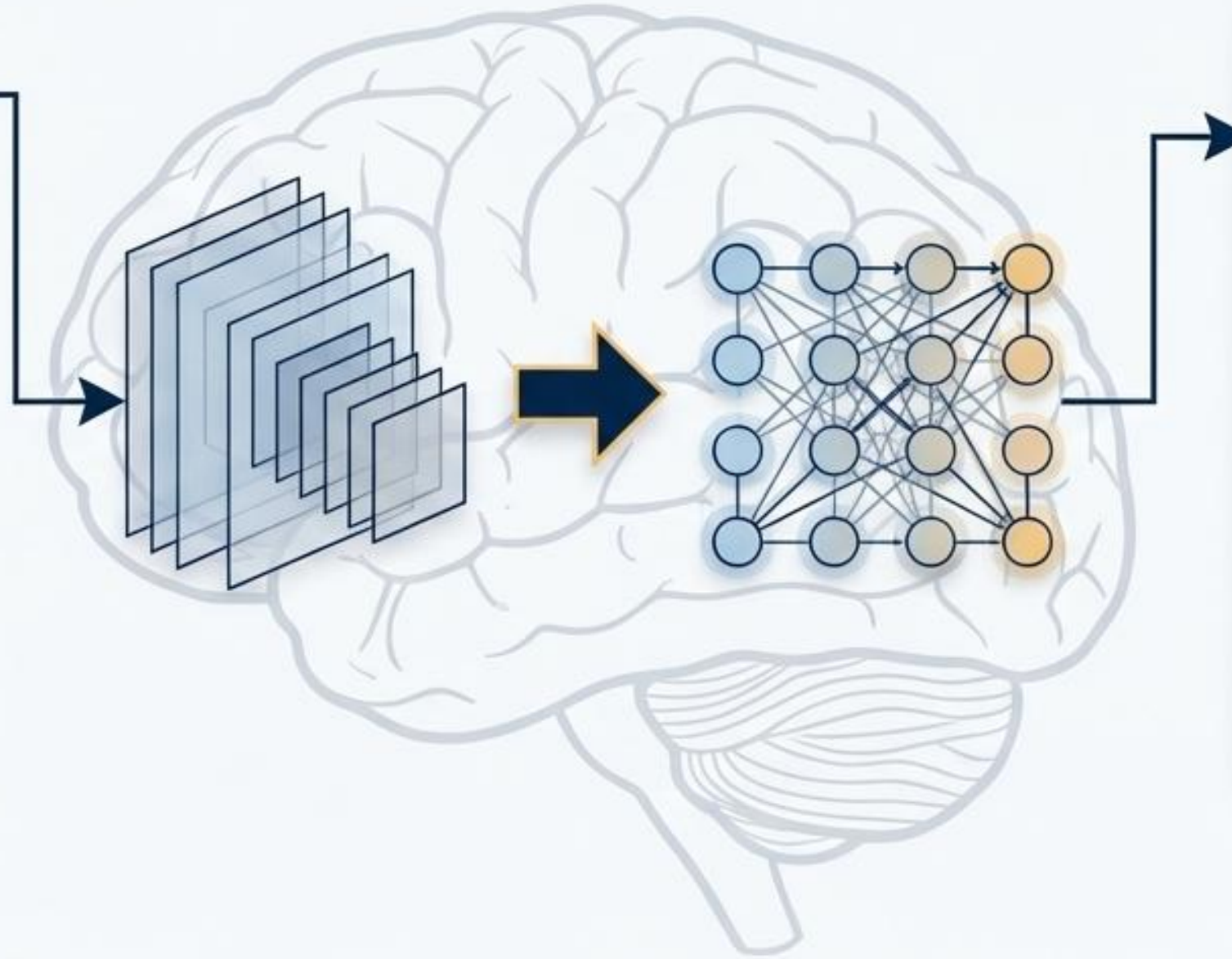


# Core Architectural Innovation: The Hybrid Model

## The Eyes: 3D CNN Encoder

Extracts local textures (e.g., connucter)

Extracts local textures (e.g., ragged tumor borders) using 3D convolutional kernels (3x3x3) and ResNet-style skip-connections.

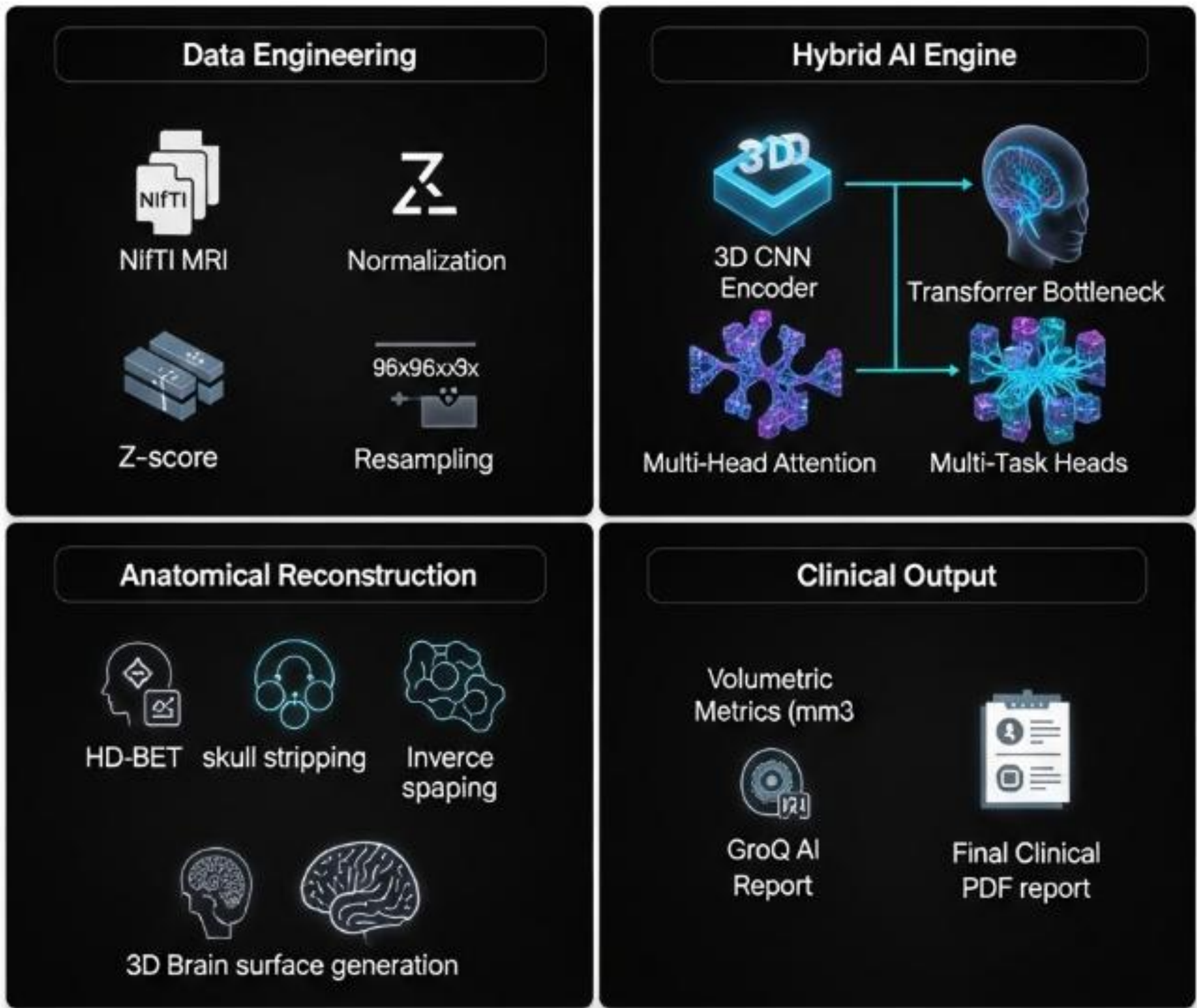


## The Brain: Transformer Bottleneck

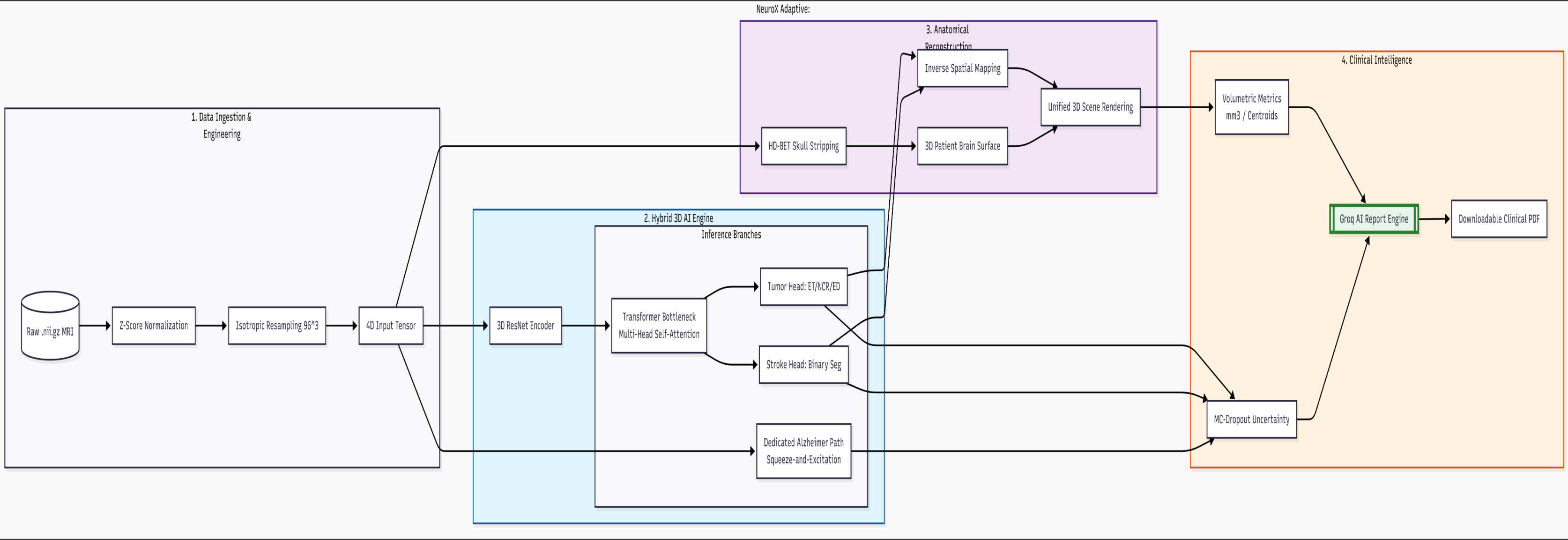
Maps global context. Employs Multi-Head Self-Attention and precise spatial encoding to understand long-range anatomical relationships across hemispheres.



# NeuroX Adaptive: Global Functional Architecture

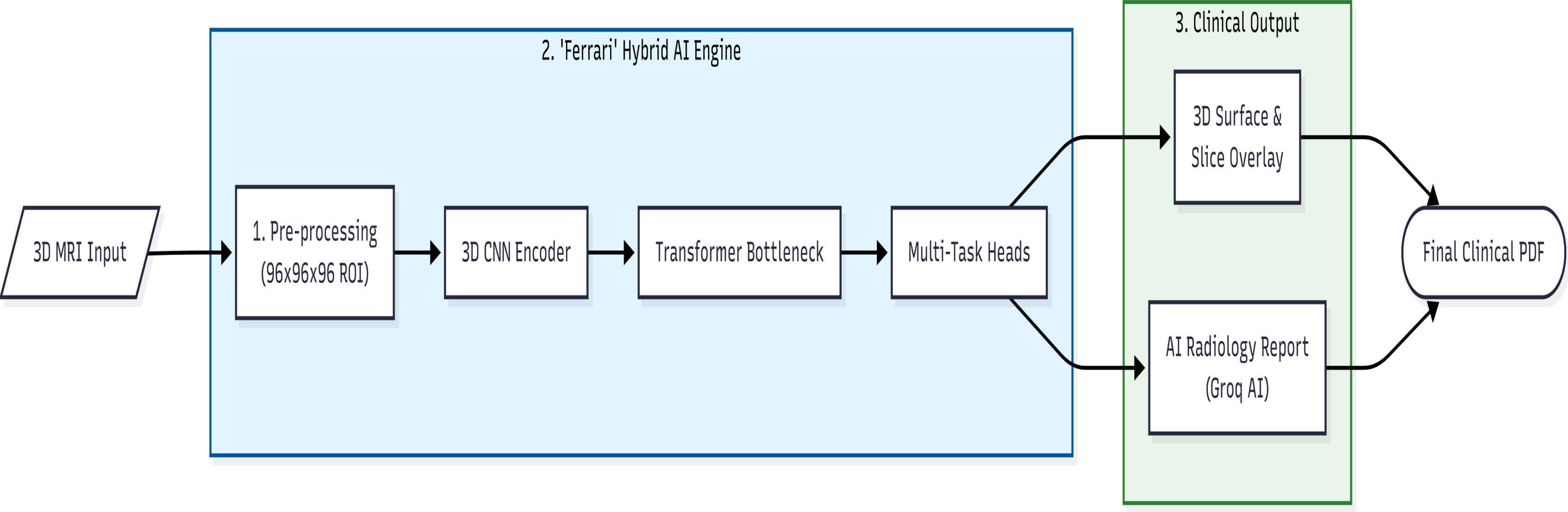


# NeuroX Adaptive: Architecture Diagram

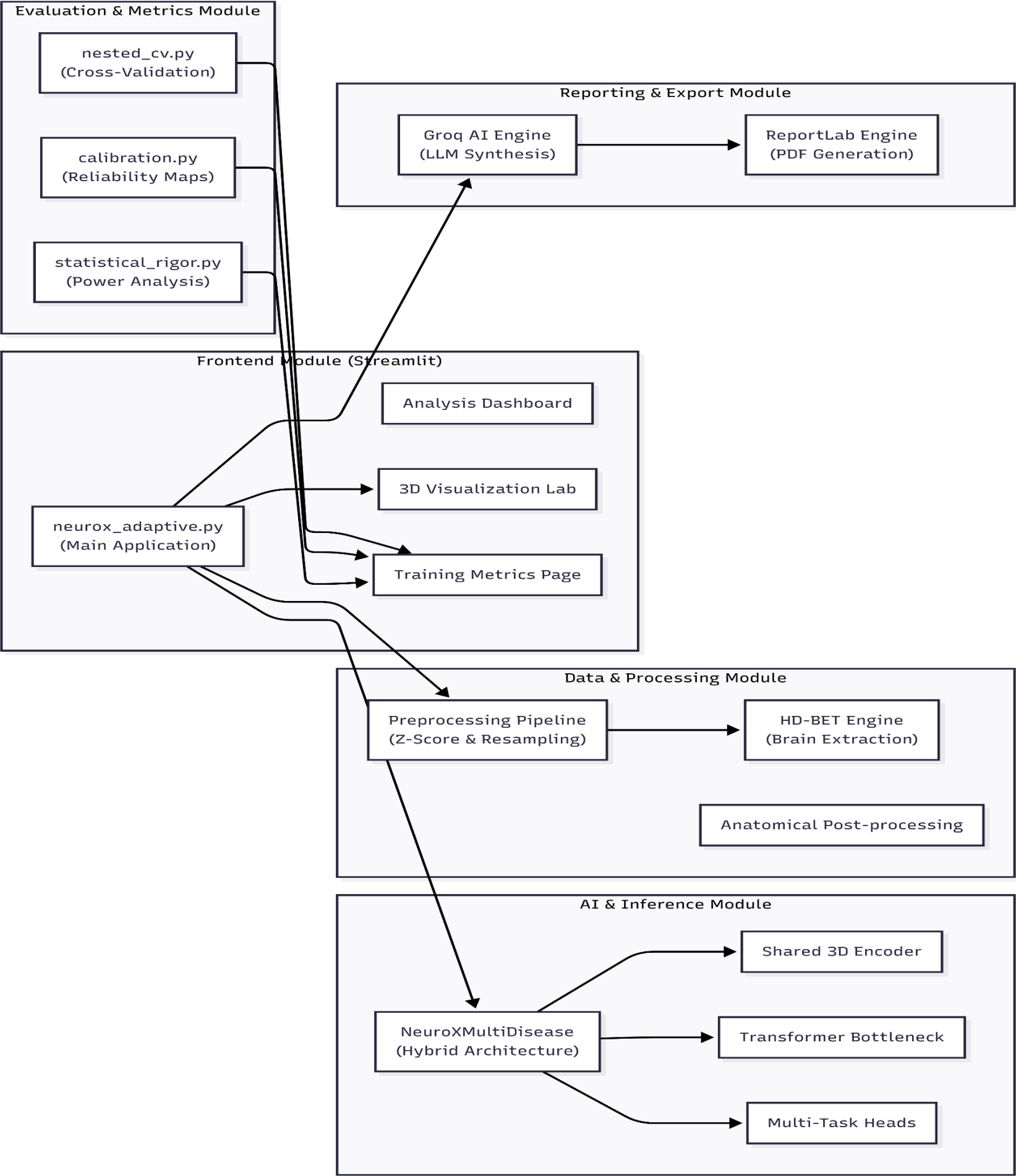




# NeuroX Adaptive: Flow Chat

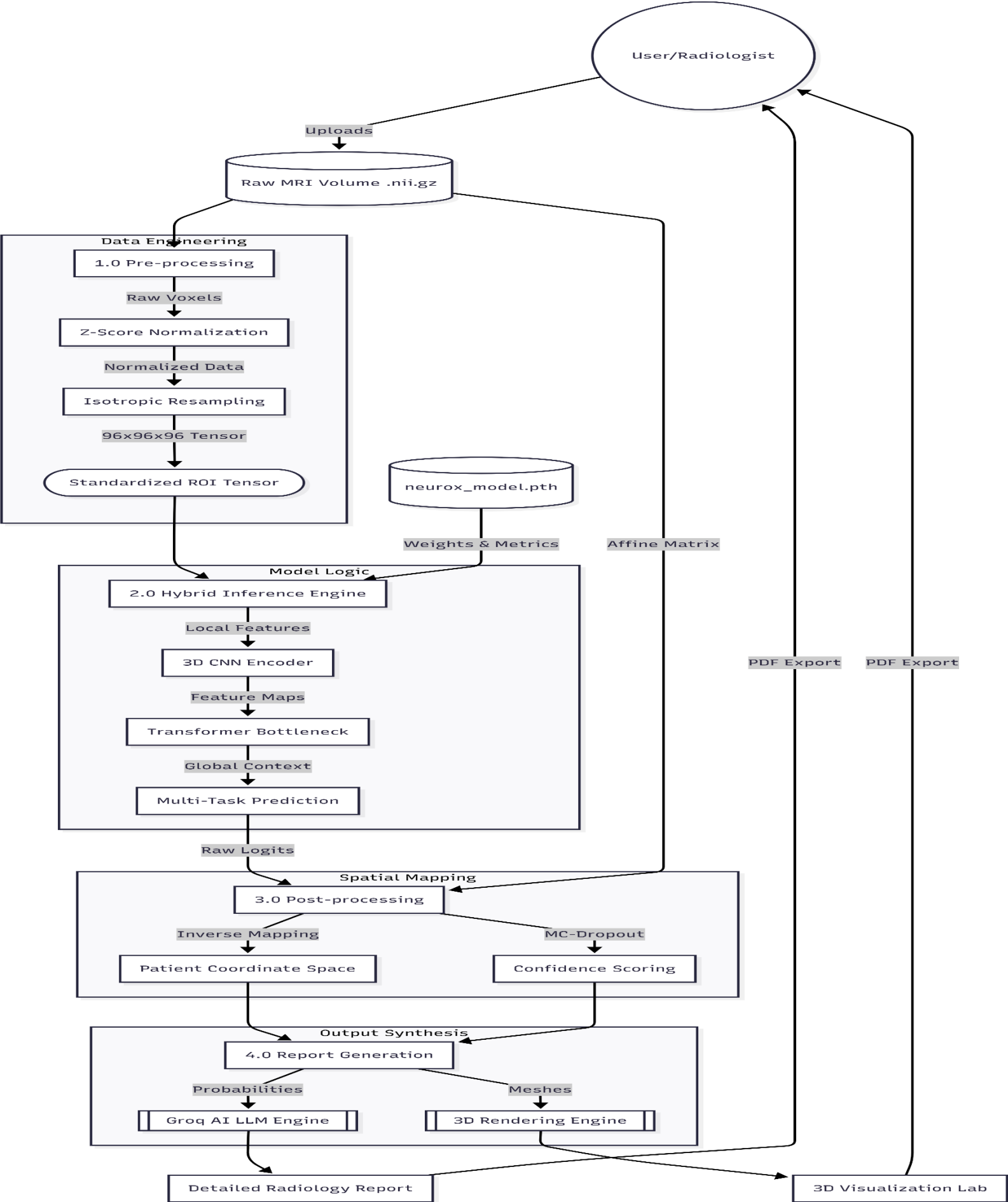


# NeuroX Adaptive: Modal Diagram

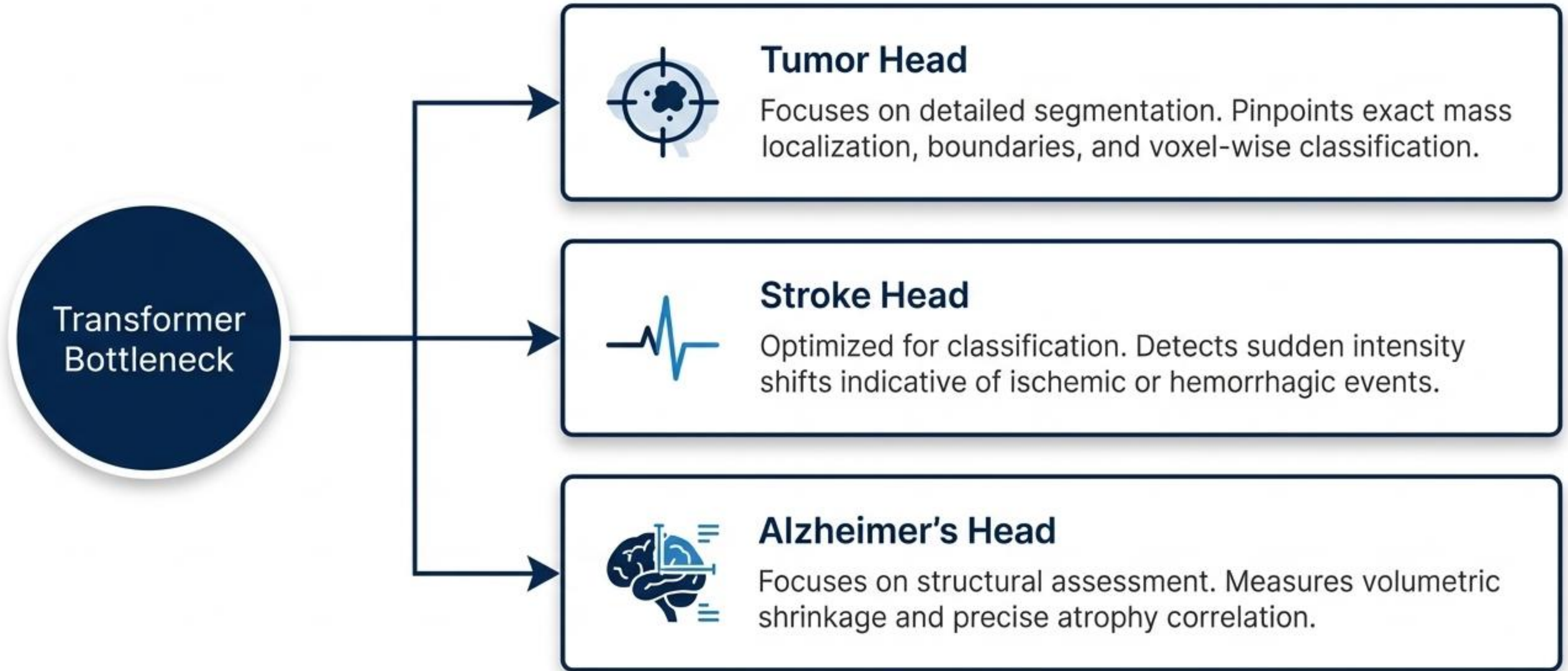




# NeuroX Adaptive: Data Flow Diagram

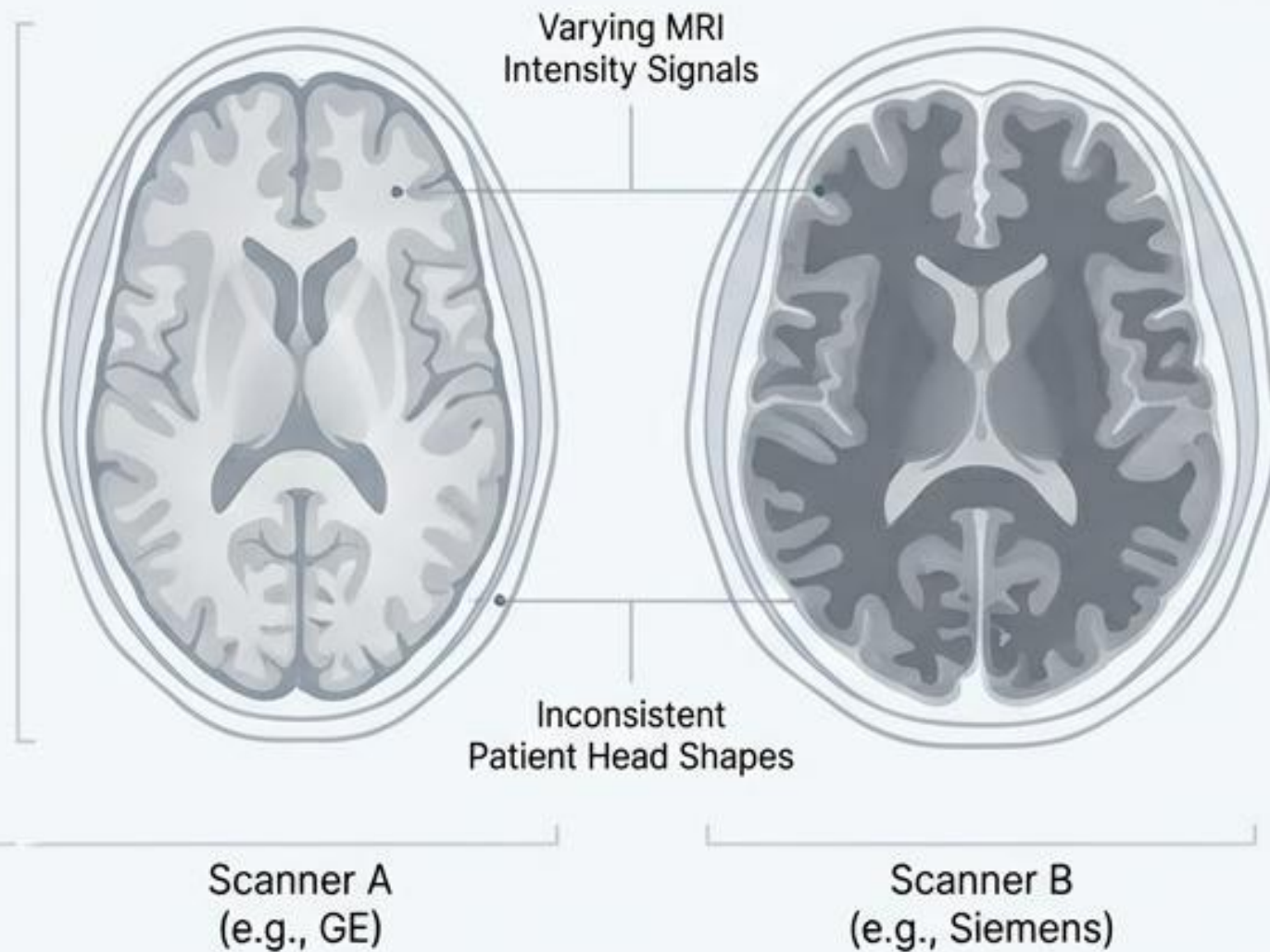


# Multi-Task Decoupled Heads

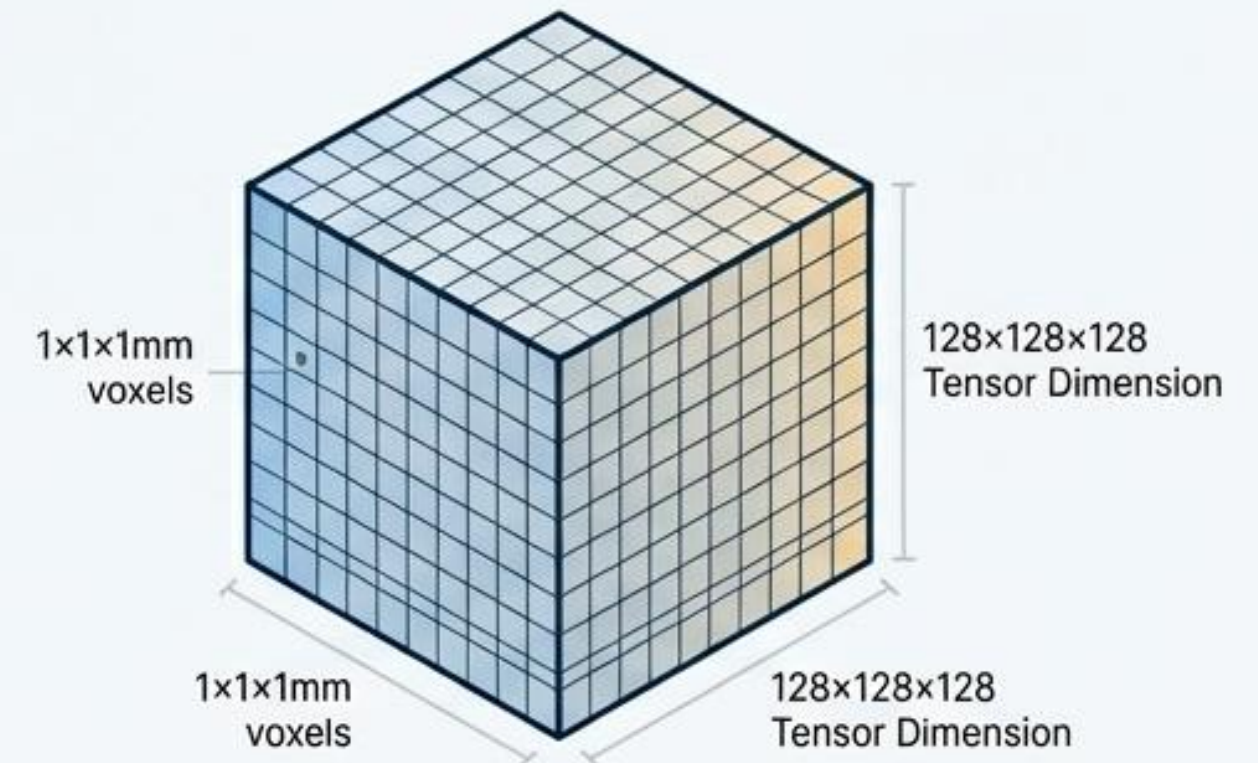


# Data Engineering: The NIfTI Processing Pipeline

## Raw Hardware Variance



## Standardized Isotropic Grid



1. **Intensity Normalization:** Z-score & Min-Max Scaling standardizes white matter values across all hardware variations.
2. **Spatial Isotropic Resampling:** Forces a universal 1x1x1mm voxel grid, strictly cropped/padded to exactly 128x128x128 tensors.

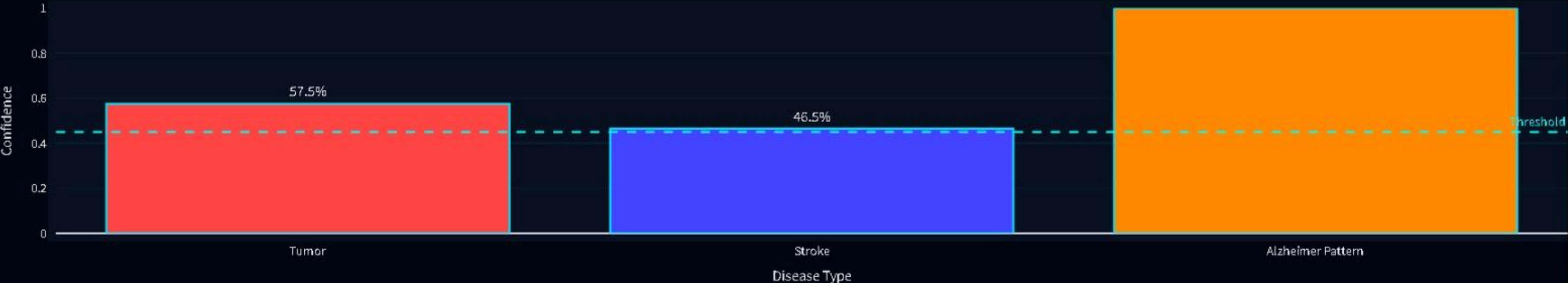


# Result:

## Clinical Interpretation

- Confidence: 100.0% - Model's belief that alzheimer pattern is present
- Uncertainty: 0.000 - Model's uncertainty about this prediction
- Status: DETECTED - Based on 45% threshold
- Recommendation: Expert review recommended for confirmation

## Disease Presence Confidence



# Result:

## 🚨 Detected Pathologies

### Tumor

Confidence: 57.5%

- 🔪 Volume: 12,912.6 mm<sup>3</sup> (12.91 mL)
- 📍 Centroid: (56.6, -60.2, -50.6) mm
- 📏 Bounding Box (Min): (31.1, -91.7, -65.2) mm
- 📏 Bounding Box (Max): (103.0, -38.1, 6.9) mm

### Stroke

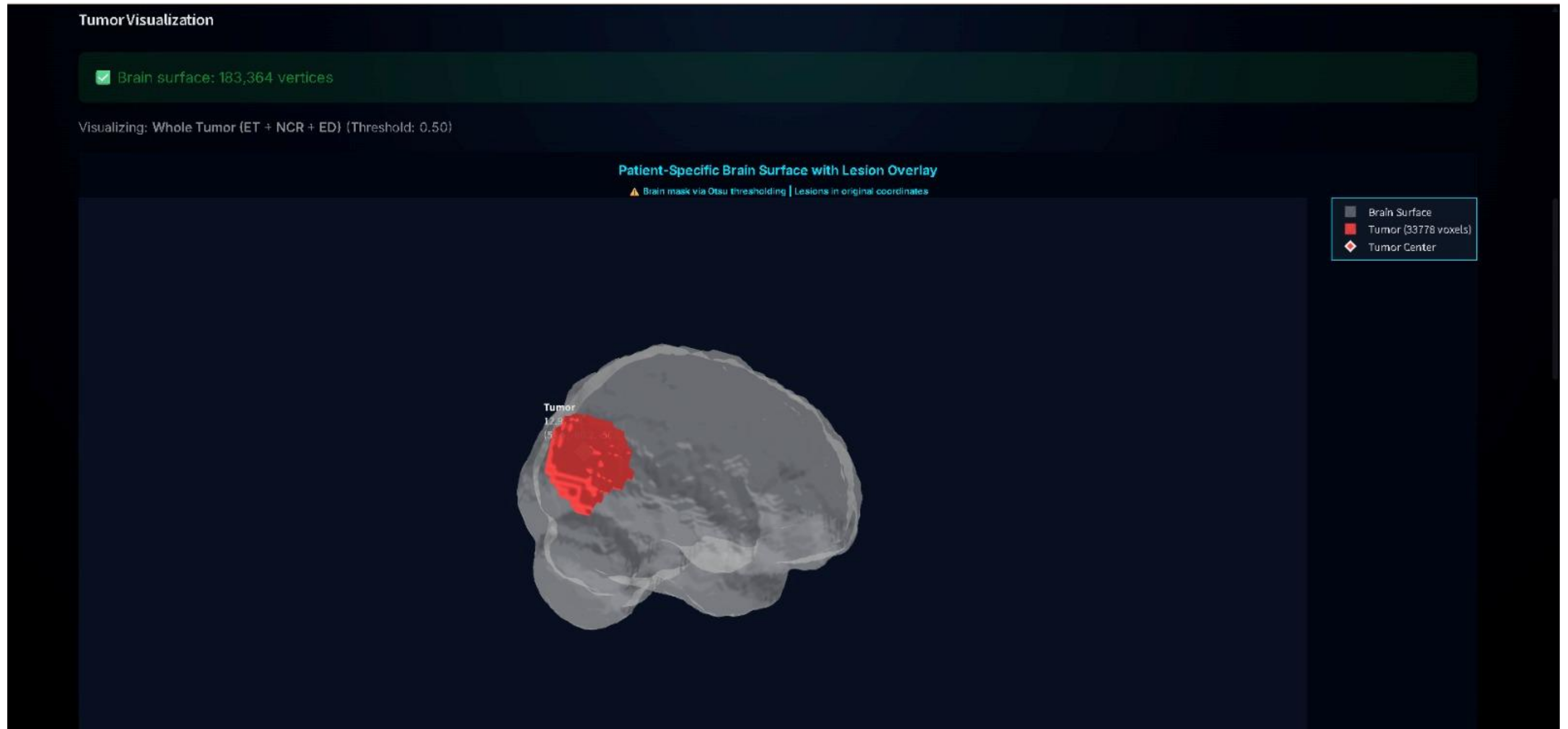
Confidence: 46.5%

- 🔪 Volume: 22,116.2 mm<sup>3</sup> (22.12 mL)
- 📍 Centroid: (55.5, -20.2, -37.4) mm
- 📏 Bounding Box (Min): (32.4, -55.6, -59.8) mm
- 📏 Bounding Box (Max): (100.9, 34.3, 7.3) mm

### Alzheimer Pattern

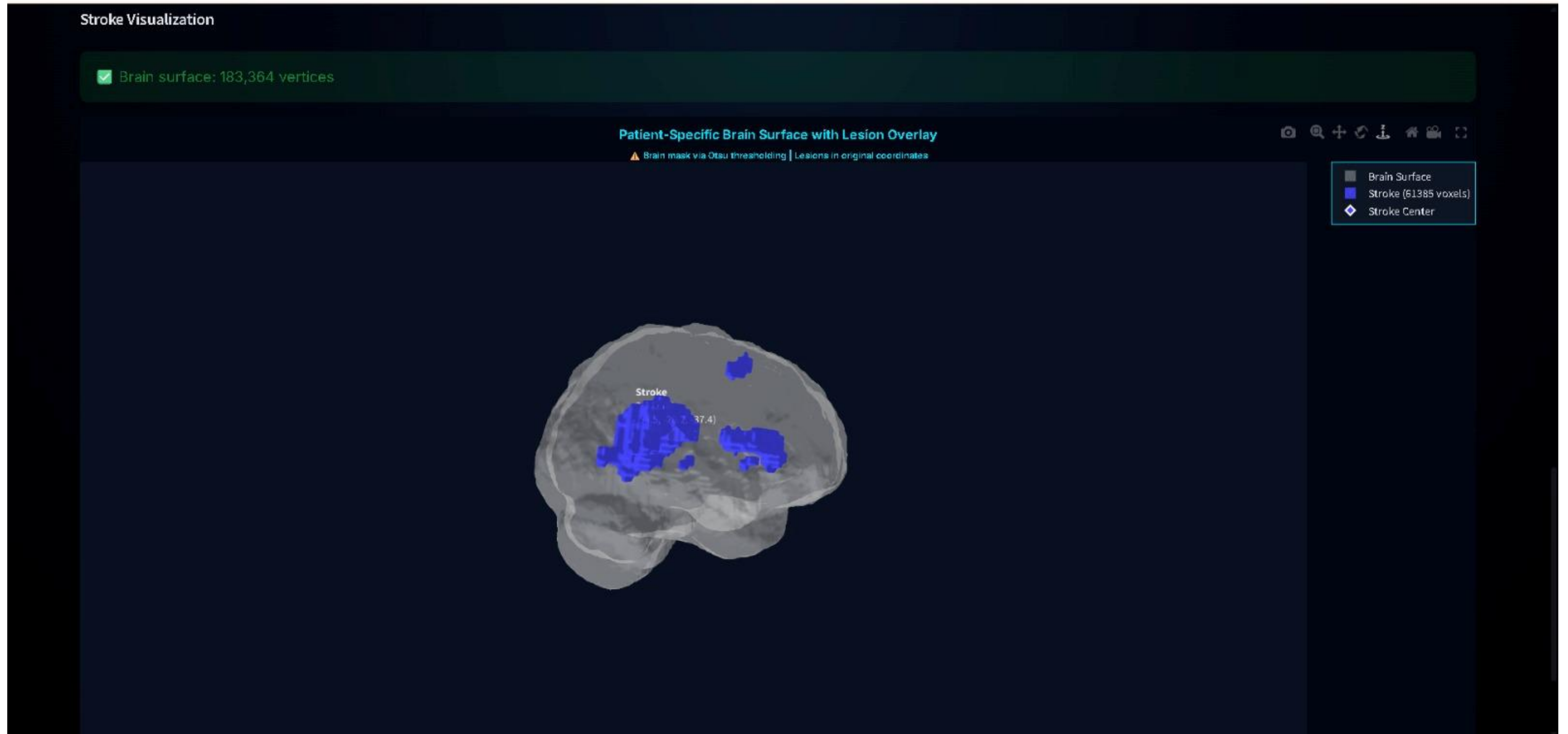
Confidence: 100.0%

# Result:





# Result:



# Result:



# Result:





# Result:

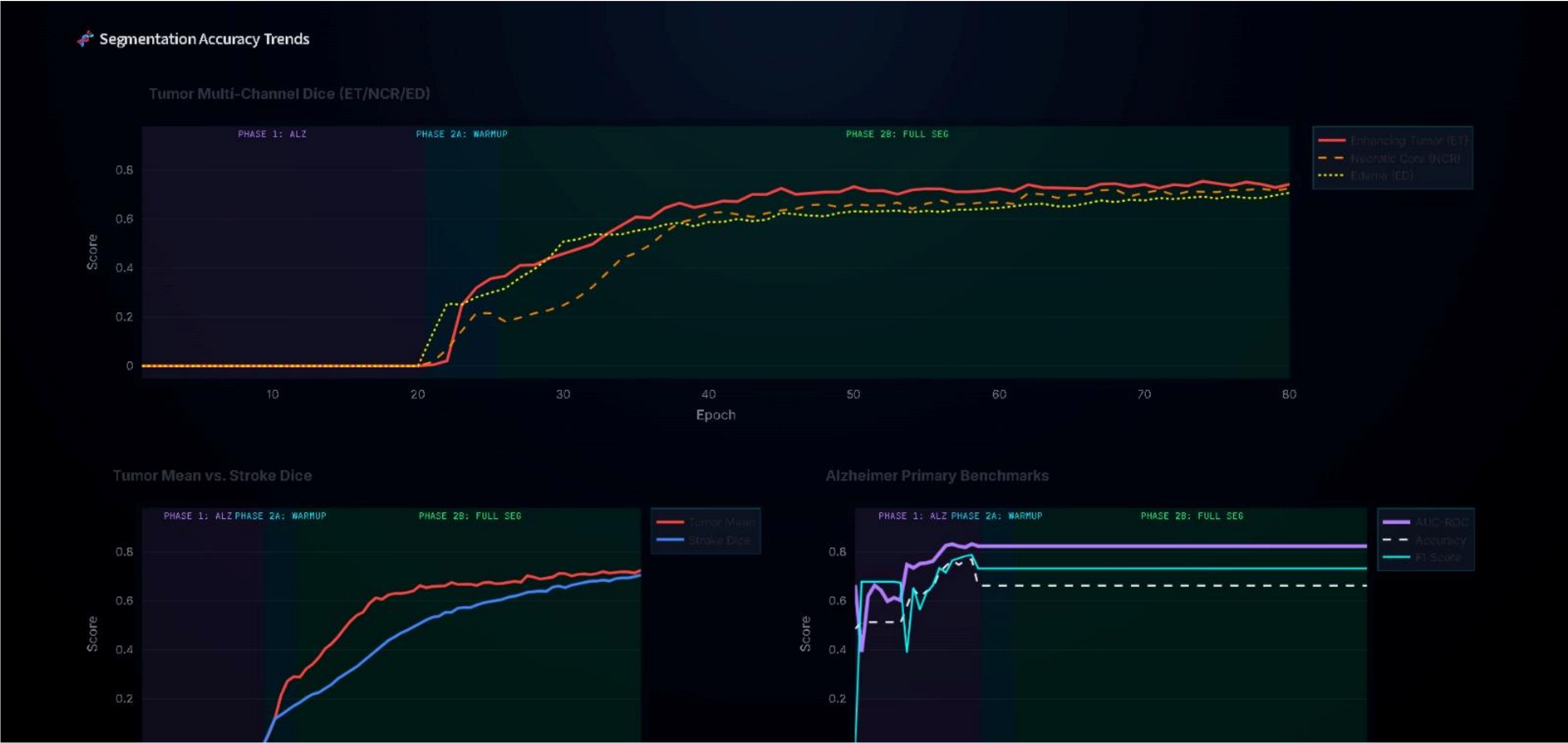
## 🏆 Detection Rigor (Alzheimer)



### 💡 Advanced Curriculum Analysis

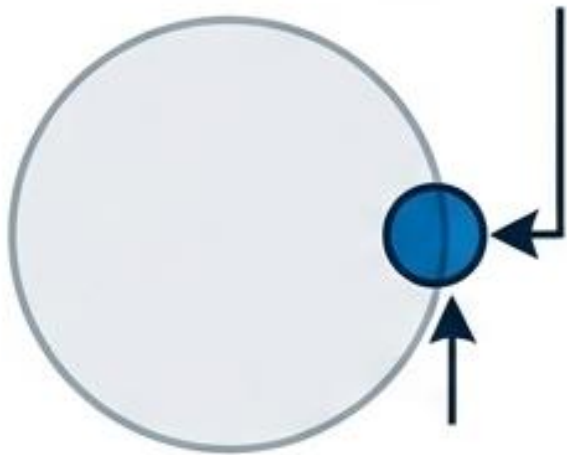
- Phase 1 (Epochs 1-20): Alzheimer Pre-training. Dedicated AlzheimerEncoder establishes base features.
- Phase 2A (Epochs 21-25): Segmentation Warmup. Convolutions are trained with lower LR to stabilize spatial paths.
- Phase 2B (Epochs 26-80): Full Segmentation. Transformer bottleneck unfrozen for global multi-disease context modeling.

# Result:



# Algorithmic Implementations (Optimization & Safety)

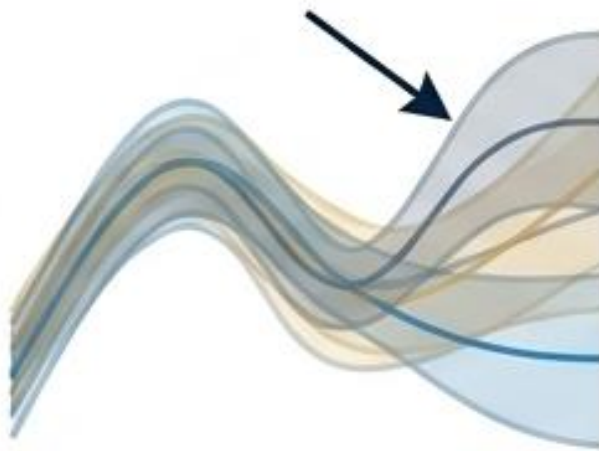
## Dice Loss Optimization



## Overcoming Class Imbalance (Dice Loss)

A brain tumor may represent only 0.5% of total volume. Dice Loss actively ignores empty background space, penalizing the model strictly on exact prediction vs. actual overlap.

## Monte-Carlo Dropout

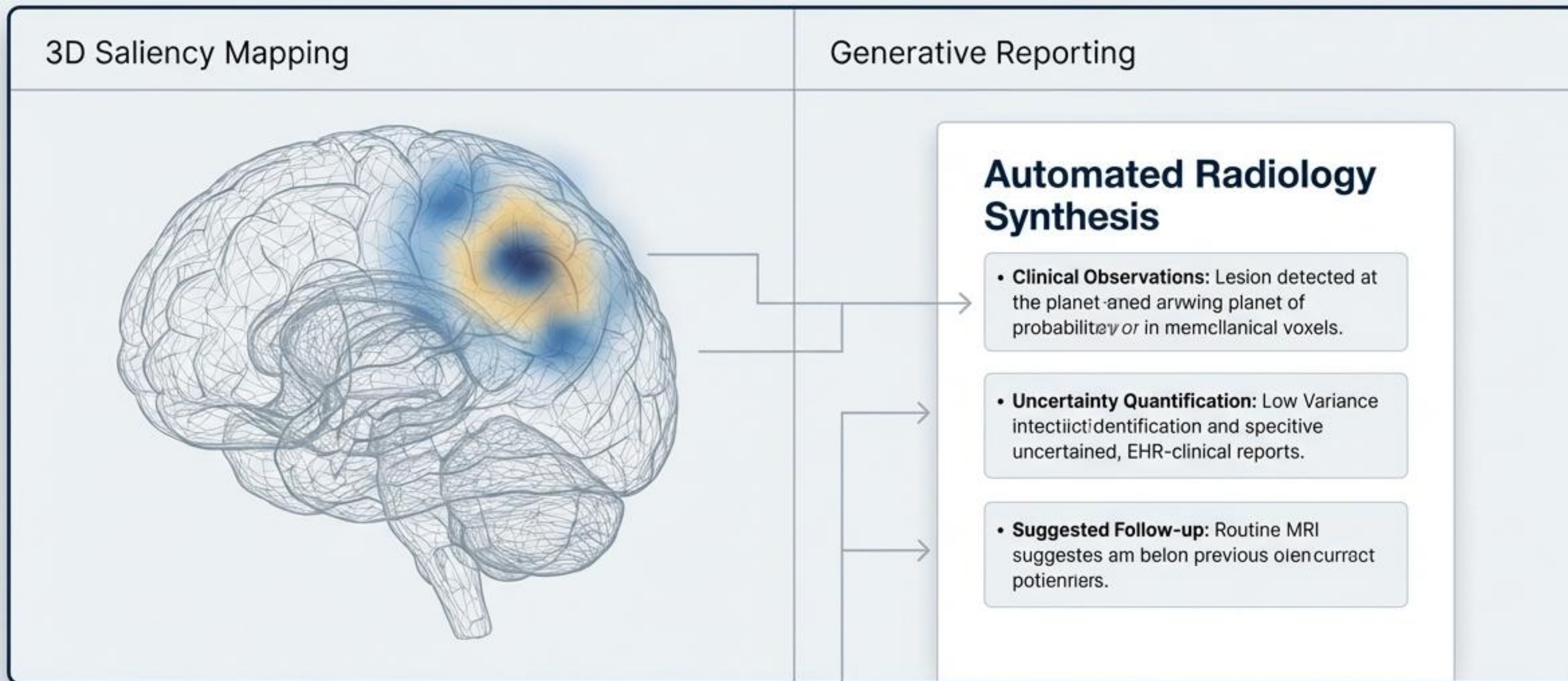


## Bayesian Clinical Safety (MC Dropout)

The network runs inference 10x with active dropout. High variance across iterations automatically triggers a "High Uncertainty" flag for manual audit. "I don't know" is a necessary medical output.



# Explainability & Generative Reporting Modules



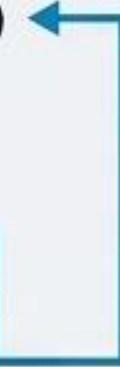
**3D Grad-CAM:** Backpropagates decisions to input voxels to generate precise 3D saliency maps, eliminating the Black Box.

**Groq AI Integration:** Llama-3 (70B) synthesizes raw probability scores into structured, EHR-ready clinical reports.

# Code Architecture (Core Snippets)

## models/hybrid\_architecture.py

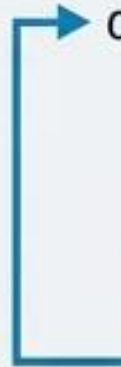
```
class Hybrid3DModel(nn.Module):  
    self.encoder = CNN3DEncoder()  
    self.bottleneck = TransformerBottleneck()
```



**Architectural Handoff:** The exact junction where local CNN textures are converted into global context.

## neurox\_adaptive.py

```
def predict(nifti_path):  
    tensor = load_and_scale(nifti_path, size=128)  
    out_tumor, out_stroke, out_alz = model(tensor)
```



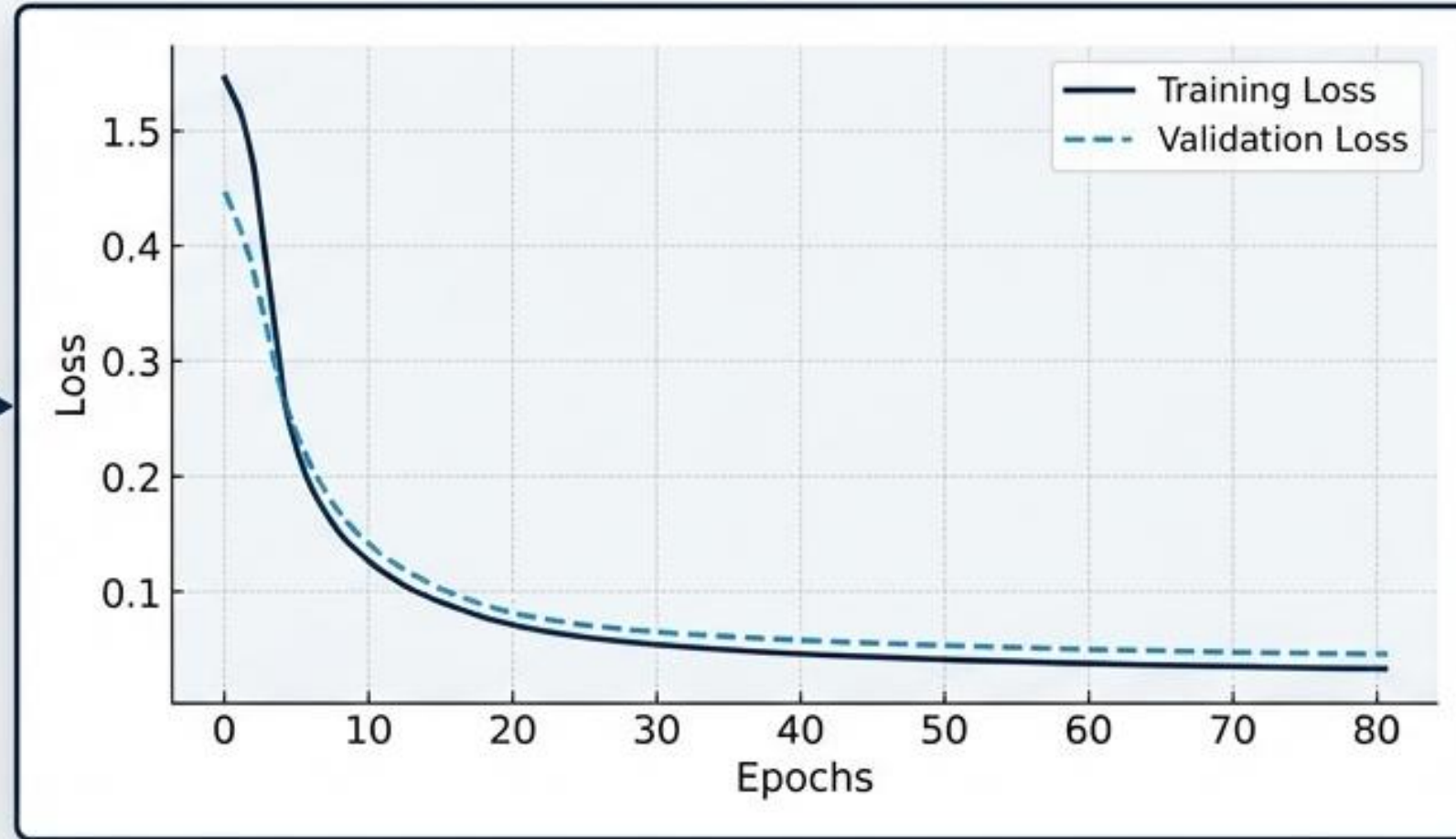
**Inference Pipeline:** unified  $128^3$  tensor routed simultaneously to the multi-task decoupled heads.



# Training Rigor & Convergence

## Hardware & Setup

Tesla T4/P100 GPUs.  
AdamW optimizer with  
dynamic learning rate  
scheduling (starts  $1e-4$ ).



## Convergence Velocity

Rapid loss decline in the  
first 20 epochs indicates  
strong foundational  
architecture.

## Validation Rigor

Minimal gap between train/val curves indicates  
negligible overfitting, verified via Nested Cross-Validation.



# Clinical Performance Metrics



## Tumor Detection

~0.84 Dice Score indicates exceptionally high segmentation boundary overlap.



## Stroke Identification

High precision/recall, identifying micro-lesions as small as  $2\text{mm}^3$ .



## Alzheimer's Staging

Strong correlation between predicted ventricle atrophy and clinical stages.



## Model Calibration

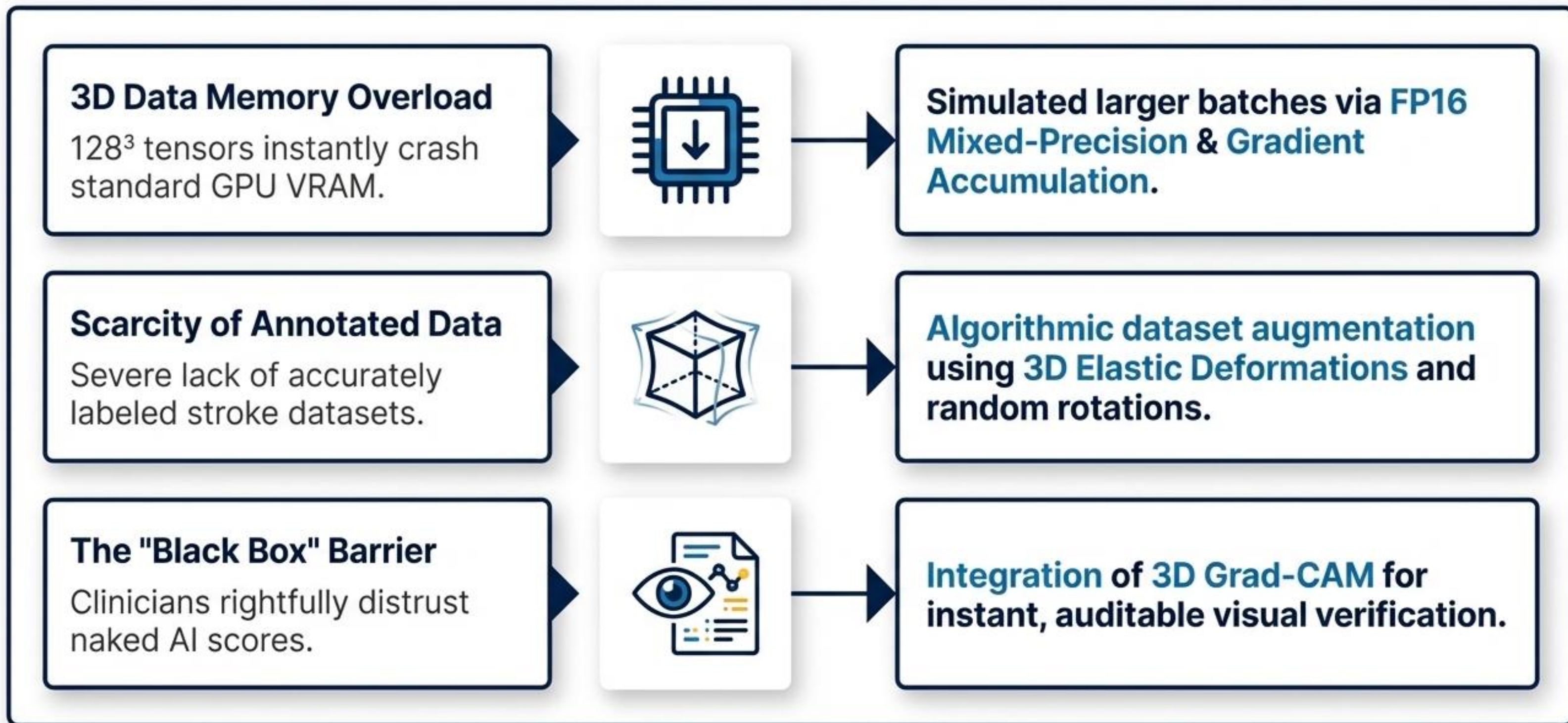
An 80% AI confidence score mathematically correlates to 80% real-world accuracy.

# System Benchmarks vs. Baseline

|                       | Standard 3D-CNN                   | NeuroX Adaptive Hybrid                                                    |
|-----------------------|-----------------------------------|---------------------------------------------------------------------------|
| Diagnostic Precision  | Baseline False Positives          | ✓ 14% Reduction in False Positives via global context awareness.          |
| Hardware Efficiency   | Standard FP32 Memory Load         | ✓ 60% Inference Time Reduction using FP16 Mixed-Precision.                |
| Throughput Capability | Variable processing time per scan | ✓ 12 Volumetric Scans per Minute fully processed on a single 16GB T4 GPU. |

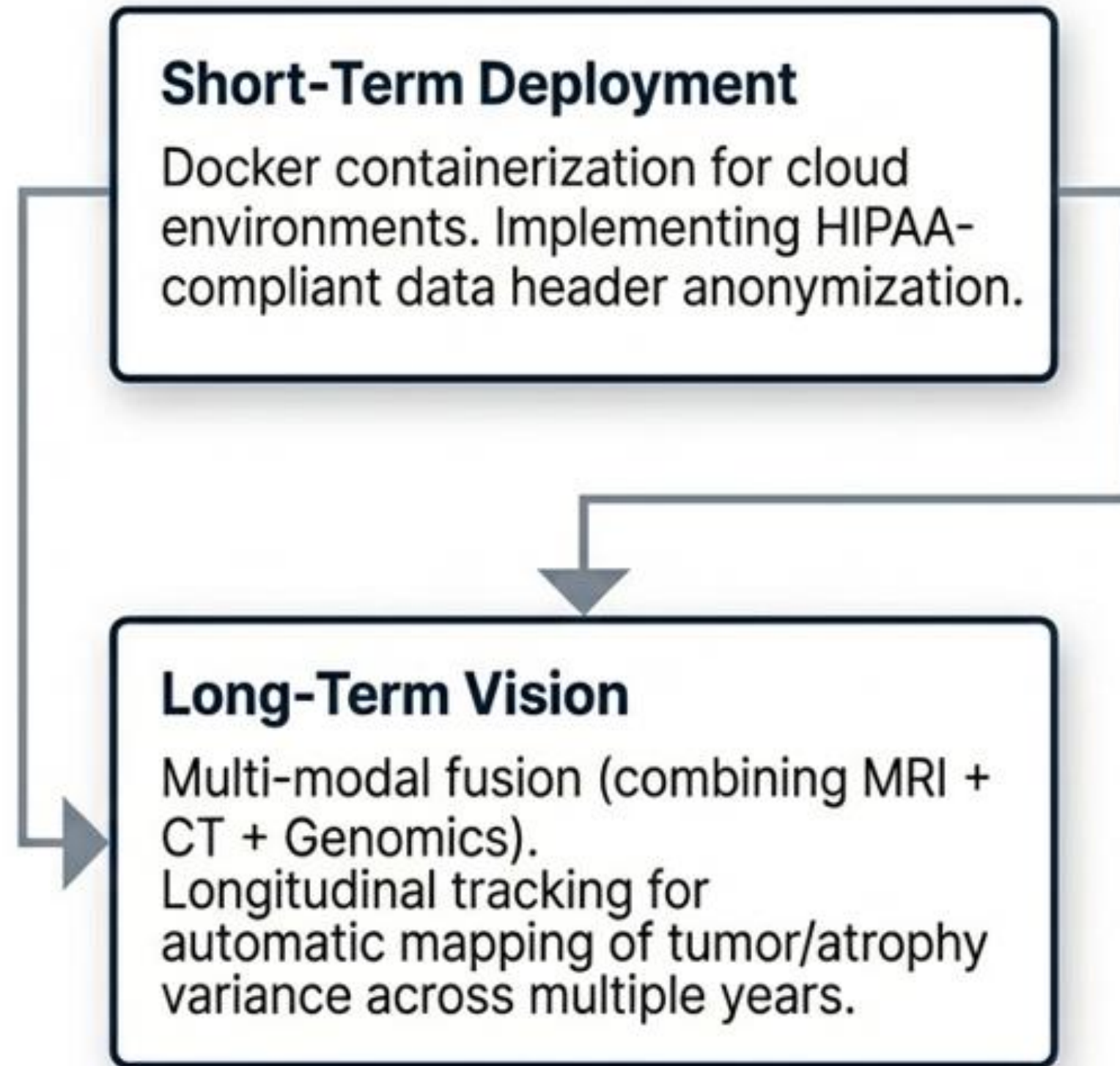


# Engineering the Impossible: Challenges Overcome





# Future Scope:



# References:

| # | Reference (APA 7th Edition)                                                                                                                                                                                                 | Core Relevance to Project                                                                                                                    |
|---|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | <b>Gal, Y., &amp; Ghahramani, Z. (2016).</b> Dropout as a Bayesian approximation: Representing model uncertainty in deep learning. <i>Proceedings of the 33rd International Conference on Machine Learning</i> , 1050–1059. | The mathematical foundation for the <b>Monte-Carlo Dropout</b> used in your <code>PresenceHead</code> to quantify diagnostic uncertainty.    |
| 2 | <b>Isensee, F., et al. (2019).</b> Automated brain extraction of multisequence MRI using artificial neural networks. <i>Human Brain Mapping</i> , 40(17), 4952–4964.                                                        | The primary reference for <b>HD-BET</b> , the gold-standard tool your system uses for medical-grade skull stripping.                         |
| 3 | <b>Vaswani, A., et al. (2017).</b> Attention is all you need. <i>Advances in Neural Information Processing Systems</i> , 5998–6008.                                                                                         | Provides the <b>Transformer Self-Attention</b> logic implemented in your <code>TransformerBottleneck3D</code> for global anatomical context. |
| 4 | <b>Ronneberger, O., Fischer, P., &amp; Brox, T. (2015).</b> U-Net: Convolutional networks for biomedical image segmentation. <i>MICCAI 2015</i> , 234–241.                                                                  | The foundational architecture for your <b>SegmentationDecoder</b> , which identifies tumor and stroke boundaries.                            |
| 5 | <b>Hu, J., Shen, L., &amp; Sun, G. (2018).</b> Squeeze-and-excitation networks. <i>Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition</i> , 7132–7141.                                           | Basis for the <b>SEBlock</b> in your <code>AlzheimerEncoder</code> that helps focus the model on relevant channel features.                  |

# References:

| #  | Reference (APA 7th Edition)                                                                                                                                                       | Core Relevance to Project                                                                                              |
|----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|
| 6  | <b>Guo, C., et al. (2017).</b> On calibration of modern neural networks. <i>Proceedings of the 34th International Conference on Machine Learning</i> , 1321–1330.                 | Justifies the use of <b>Calibration Curves</b> to ensure your model's confidence scores match clinical reality.        |
| 7  | <b>Sechidis, K., Tsoumakas, G., &amp; Vlahavas, I. (2011).</b> On the stratification of multi-label data. <i>Machine Learning and Knowledge Discovery in Databases</i> , 145–158. | Supports the <b>Multi-Label</b> approach where one patient can be diagnosed with multiple diseases simultaneously.     |
| 8  | <b>Vickers, A. J., &amp; Elkin, E. B. (2006).</b> Decision curve analysis: A novel method for evaluating prediction models. <i>Medical Decision Making</i> , 26(6), 565–574.      | The standard for the <b>Decision Curve Analysis</b> found in your <code>statistical_rigor.py</code> evaluation module. |
| 9  | <b>Varma, S., &amp; Simon, R. (2006).</b> Bias in error estimation when using cross-validation for model selection. <i>BMC Bioinformatics</i> , 7(1), 1–8.                        | Provides the methodology for the <b>Nested Cross-Validation (5x2)</b> used to validate your model's performance.       |
| 10 | <b>Lin, T. Y., et al. (2017).</b> Focal loss for dense object detection. <i>Proceedings of the IEEE International Conference on Computer Vision</i> , 2980–2988.                  | Supports the loss optimization strategies used to handle the <b>class imbalance</b> of small brain lesions.            |